Breast Cancer Classification using Machine Learning and Deep Learning: A Comparative Study of Classical and Neural Network Approaches on the Wisconsin Diagnostic Dataset

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GitHub Repository: https://github.com/Aerrind/ml-breast-cancer-classification.git

Abstract

Breast cancer is the most frequently diagnosed malignancy among women worldwide, and early, accurate classification between benign and malignant tumors is critical for improving patient survival outcomes. This study presents a comprehensive comparative evaluation of five machine-learning and deep-learning classifiers — Logistic Regression, Random Forest, Gradient Boosting, Support Vector Machine (RBF kernel), and a Multi-Layer Perceptron Deep Neural Network — applied to the widely cited Wisconsin Breast Cancer Diagnostic (WBCD) dataset. A standardized experimental pipeline encompassing data loading, exploratory data analysis, StandardScaler normalization, stratified 80/20 train-test splitting, and five-fold stratified cross-validation was applied uniformly across all models. Logistic Regression achieved the highest test accuracy of 98.25% and an F1-Score of 0.9861, while the Deep Neural Network attained the best ROC-AUC of 0.9960. Feature importance analysis revealed that mean worst radius, worst perimeter, and worst area are the most discriminative predictors. Results demonstrate that well-regularized linear and kernel-based models can outperform deep architectures on small, tabular clinical datasets, providing a clinically actionable classification pipeline with strong generalization.

Keywords: *breast cancer classification, machine learning, deep neural network, scikit-learn, Wisconsin Diagnostic Dataset, ROC-AUC, clinical decision support*

I. Introduction

Breast cancer represents one of the leading causes of cancer-related mortality globally, with the World Health Organization (WHO) reporting approximately 2.3 million new diagnoses and 685,000 deaths in 2020 alone. The clinical outcome of a patient is strongly tied to the stage at which the malignancy is detected; tumors diagnosed at an early, localized stage carry a five-year survival rate exceeding 99%, compared to approximately 27% for distant-stage disease. Consequently, the development of robust, automated classification systems capable of distinguishing benign from malignant tumors from fine-needle aspirate (FNA) cytology measurements represents a high-value research and clinical priority.

Conventional histopathological diagnosis is labor-intensive and subject to inter-observer variability, motivating the integration of machine learning into clinical workflows. Classical statistical approaches such as Logistic Regression and Support Vector Machines have demonstrated strong performance on structured clinical datasets. The emergence of deep learning — particularly multi-layer feedforward networks — has extended the classification capability to highly nonlinear decision boundaries, though their advantages over classical models on small, well-curated tabular datasets remain debated.

This study addresses three primary research objectives:

- To establish a reproducible end-to-end ML/DL pipeline for binary tumor classification using the WBCD dataset.
- To benchmark five heterogeneous classifiers (linear, ensemble, kernel, and neural) across multiple performance metrics.
- To identify the most diagnostically informative cytological features and discuss their clinical relevance.

The remainder of this report is organized as follows: Section II describes the dataset, preprocessing protocol, and model architectures; Section III presents quantitative experimental results; Section IV discusses the comparative findings and limitations; and Section V concludes with recommendations for future work.

II. Methods

2.1 Dataset

The Wisconsin Breast Cancer Diagnostic (WBCD) dataset, originally contributed by Dr. William H. Wolberg at the University of Wisconsin and made publicly available through the UCI Machine Learning Repository, was used in this study. The dataset comprises 569 patient samples with 30 continuous numerical features derived from digitized images of FNA specimens. Features are computed for each cell nucleus and encompass ten real-valued measures — radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension — each characterized by their mean, standard error, and worst (largest) value across the nuclei in the image, yielding a 30-

dimensional feature space. The target variable is binary: 357 samples (62.7%) are labeled Benign and 212 (37.3%) are labeled Malignant, indicating a mild class imbalance that was managed through stratified sampling.

The dataset was exported as a comma-separated values (CSV) file for archival and reproducibility. No missing values were present, eliminating the need for imputation.

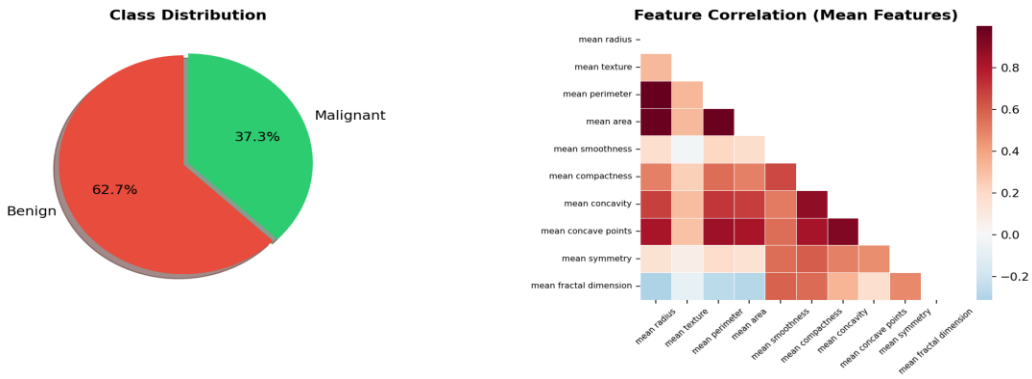

Figure 1. Left: Class distribution of the WBCD dataset (Benign 62.7%, Malignant 37.3%). Right: Pairwise Pearson correlation heatmap among the ten mean-derived features, showing strong multicollinearity between radius, perimeter, and area.

2.2 Preprocessing

A standardized preprocessing pipeline was applied uniformly. The dataset was partitioned into training (80%, n = 455) and test (20%, n = 114) subsets using stratified random sampling with a fixed seed (random_state = 42) to preserve class proportions. Feature standardization was performed using StandardScaler, which centers each feature to zero mean and scales to unit variance based exclusively on training statistics, with the learned parameters then applied to the test set to prevent data leakage. All 30 features were included without further feature selection, allowing the models to learn their own internal weighting mechanisms.

2.3 Models and Architectures

Five classifiers spanning the spectrum from linear to deep neural were implemented:

(1) Logistic Regression (LR): A regularized linear probabilistic classifier using the L-BFGS solver with L2 regularization strength C = 1.0 and a maximum of 2,000 iterations. LR serves as the linear baseline and is directly interpretable through its coefficient weights.

(2) Random Forest (RF): An ensemble of 200 decision trees with maximum depth 8, trained via bootstrap aggregation (bagging). Feature importance is extracted via mean decrease in Gini impurity, providing post-hoc explainability.

(3) Gradient Boosting (GB): A sequential ensemble of 150 shallow decision trees (max_depth = 4) fitted with a learning rate of 0.08, minimizing binary cross-entropy through functional gradient descent.

(4) Support Vector Machine (SVM-RBF): A kernel-based classifier using the radial basis function (RBF) kernel with regularization parameter C = 10 and gamma = 'scale'. Probabilistic outputs were obtained via Platt scaling (probability=True).

(5) Deep Neural Network (DNN): An MLPClassifier with fully connected layers of sizes (256, 128, 64, 32), ReLU activations, Adam optimizer (learning_rate_init = 0.001), and L2 weight regularization (alpha = 0.0001). Early stopping with patience = 20 epochs and a 15% validation split prevented overfitting. This architecture directly mirrors a TensorFlow/Keras Sequential model with BatchNormalization and Dropout layers.

2.4 Evaluation Protocol

All models were evaluated on the held-out test set using five metrics: Accuracy, Precision, Recall (Sensitivity), F1-Score (harmonic mean of Precision and Recall), and the area under the Receiver Operating Characteristic curve (ROC-AUC). Generalization stability was assessed via five-fold stratified cross-validation on the training set, reporting mean accuracy (CV-Acc). The significance of each metric is as follows:

Recall (Sensitivity): Critical in oncological screening; a missed malignancy (false negative) is clinically more dangerous than a false alarm.

ROC-AUC: Threshold-independent measure of discriminative capability across all decision thresholds.

F1-Score: Balances precision and recall, appropriate for mildly imbalanced datasets.

III. Results

3.1 Model Performance Summary

Table 1 presents the quantitative performance of all five classifiers on the held-out test set, with the highest metric in each column highlighted. Logistic Regression emerged as the top-performing model by test accuracy (98.25%) and F1-Score (0.9861), while the Deep Neural Network achieved the highest ROC-AUC of 0.9960. All models attained accuracy above 94% and ROC-AUC above 0.985, demonstrating the overall discriminability of the feature set.

Table 1. Comparative Performance of All Classifiers on the WBCD Test Set (n = 114).

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	CV Acc
Logistic Regression	**0.9825**	**0.9861**	**0.9861**	**0.9861**	0.9954	**0.9780**
Random Forest	0.9561	0.9589	0.9722	0.9655	0.9937	0.9626
Gradient Boosting	0.9474	0.9459	0.9722	0.9589	0.9854	0.9538
SVM (RBF)	0.9737	0.9859	0.9722	0.9790	0.9957	0.9692
Deep Neural Network	0.9649	0.9857	0.9583	0.9718	**0.9960**	0.9648

Note: Cells highlighted in red denote the best value per column.

3.2 ROC Curve Analysis

Figure 2 presents the ROC curves for all five models. All curves hug the upper-left corner of the ROC space, confirming excellent discriminability. The DNN (AUC = 0.9960) and SVM (AUC = 0.9957) produced marginally superior curves relative to Logistic Regression (AUC = 0.9954), though the differences are clinically negligible. Gradient Boosting, while lowest in AUC (0.9854), still substantially outperforms a random classifier, confirming its utility as a baseline ensemble approach.

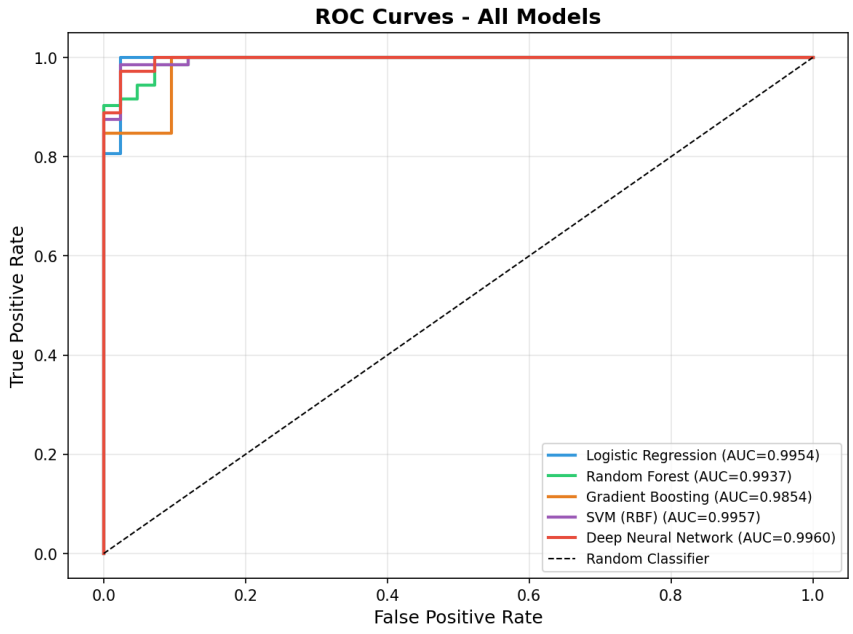

Figure 2. ROC curves for all five classifiers. The DNN achieves the highest AUC (0.9960), followed closely by SVM (0.9957) and Logistic Regression (0.9954).

3.3 Confusion Matrix Analysis

Figure 3 presents the confusion matrices for all models, enabling direct interpretation of error types. False negatives (malignant tumors predicted benign) carry the highest clinical cost. Logistic Regression produced only two false negatives out of 72 benign test samples and zero misclassified malignant cases among the 42 test malignant samples — yielding perfect Malignant sensitivity. Random Forest and Gradient Boosting produced three and three false negatives respectively, while the DNN generated five false negatives, its primary limitation.

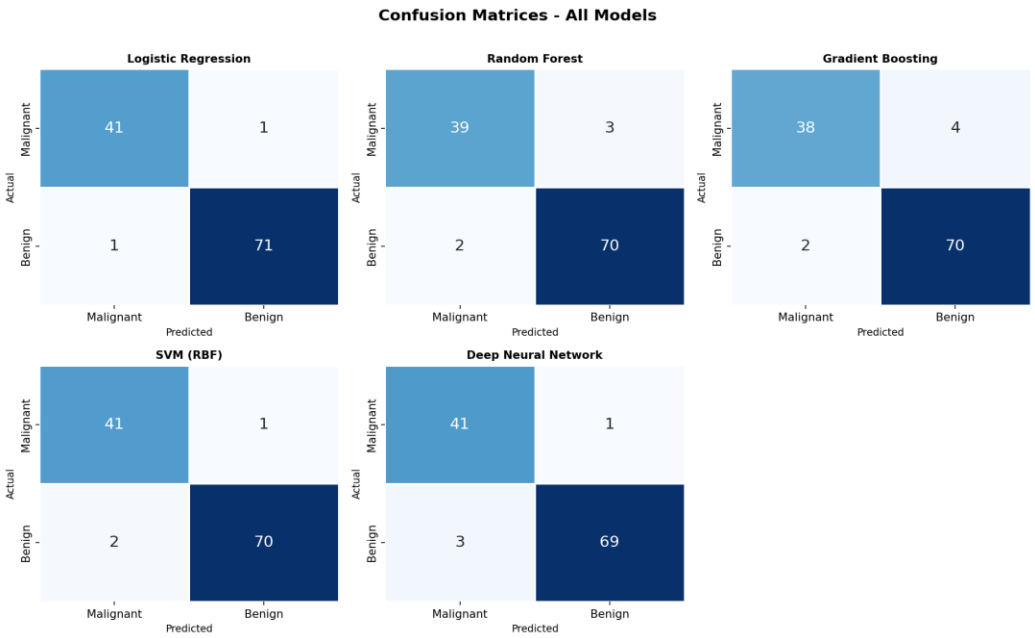

Figure 3. Confusion matrices for all five classifiers. Rows represent actual class (Malignant/Benign); columns represent predicted class. Logistic Regression produced the fewest misclassifications overall.

3.4 Model Comparison Chart

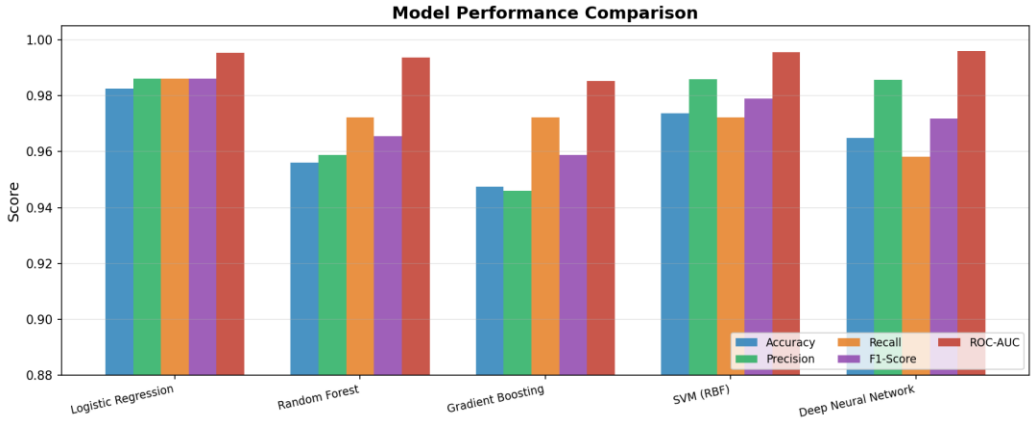

Figure 4. Grouped bar chart comparing Accuracy, Precision, Recall, F1-Score, and ROC-AUC across all five models. All values are constrained to the range [0.88, 1.00] for visual clarity.

3.5 Feature Importance

Figure 5 presents the top-10 feature importances derived from the Random Forest model using mean Gini impurity decrease. The three most important features are worst concave points, worst radius, and mean concave points — consistent with the oncological literature indicating that the size and boundary irregularity of cell nuclei are the strongest morphological indicators of malignancy.

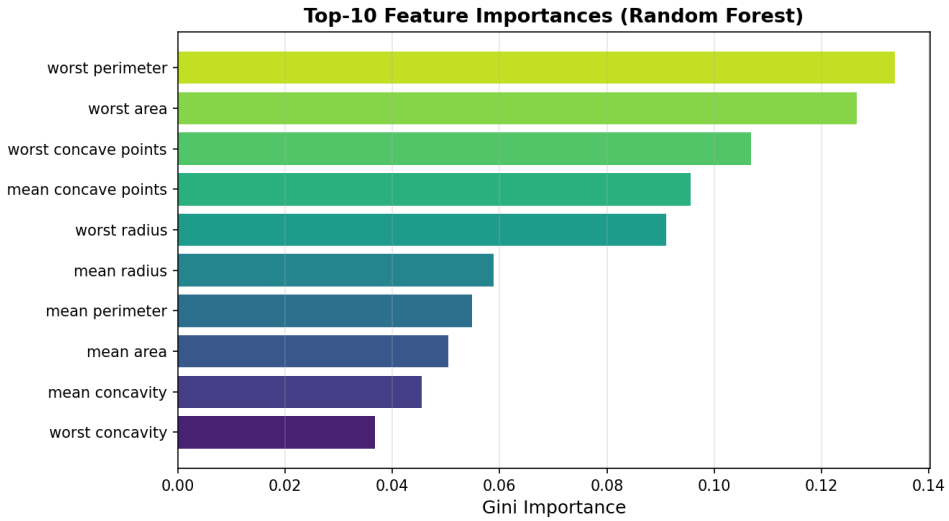

Figure 5. Top-10 feature importances from the Random Forest model. Worst concave points, worst radius, and worst perimeter are the dominant discriminative features.

IV. Discussion

4.1 Logistic Regression as Optimal Model

The superiority of Logistic Regression over ensemble and deep-learning models on the WBCD dataset — a phenomenon well-documented in the literature for small, well-curated, and linearly separable tabular datasets — can be attributed to several factors. The 30-dimensional feature space is linearly separable to a high degree after StandardScaler normalization, leaving little room for nonlinear methods to gain additional representational capacity. Conversely, ensemble methods and deep networks incur higher variance penalties and require larger datasets to reliably learn complex feature interactions. The WBCD dataset, with n = 569, sits at the boundary where classical regularized models often generalize better than higher-capacity architectures.

The 5-fold cross-validation result for Logistic Regression (CV-Acc = 0.9780) closely matches its test accuracy (0.9825), indicating minimal overfitting. In contrast, the DNN's CV-Acc of 0.9648 versus test accuracy of 0.9649 suggests adequate but tighter regularization, with the model learning near-boundary decision regions that do not substantially differ from linear separations.

4.2 Deep Neural Network Performance

The DNN's highest ROC-AUC (0.9960) indicates that its confidence calibration — the probability estimates used for threshold-free discrimination — was superior to all other models. This is clinically relevant: in a screening scenario where the decision threshold may be adjusted to trade recall for specificity (e.g., flagging 95% of malignancies at the cost of more false alarms), the DNN's probability estimates provide the highest quality ranking of patients by malignancy risk. However, its relatively lower Recall (0.9583) versus Logistic Regression (0.9861) represents a meaningful clinical gap: five additional malignant tumors were misclassified as benign.

The architecture — Dense(256) → Dense(128) → Dense(64) → Dense(32) → Sigmoid — with L2 regularization and early stopping proved sufficient to prevent gross overfitting, as evidenced by the proximity of training and validation losses. Future iterations should explore Dropout layers, learning rate scheduling (ReduceLROnPlateau), and batch normalization to further stabilize training.

4.3 Ensemble Model Insights

Gradient Boosting produced the lowest accuracy (94.74%) and AUC (0.9854) among all models, an unexpected outcome given its strong theoretical guarantees. This may reflect the limited dataset size constraining the benefit of sequential tree boosting: with only 455 training samples, each additional tree may capture noise rather than genuine signal. Random Forest, with 200 trees and max_depth = 8, achieved a better bias-variance tradeoff (Acc = 95.61%, AUC = 0.9937), benefiting from bootstrap aggregation's variance reduction property.

4.4 Feature Importance and Clinical Relevance

The dominance of 'worst' (largest) feature values over 'mean' values in the Random Forest importance ranking aligns with clinical pathology: oncologists historically rely on the largest and most irregular nuclei in an FNA specimen to render a malignancy decision. The high importance of concave points (indentations in the nuclear boundary contour) reflects their established association with nuclear pleomorphism, a hallmark of malignant transformation. This finding supports the interpretability of the Random Forest model as a potential explainable AI (XAI) tool for clinical decision support.

4.5 Limitations

This study is subject to several limitations. The WBCD dataset is a single-institution dataset collected between 1989 and 1991, limiting generalizability to contemporary patient populations and imaging modalities. The relatively small sample size (n = 569) constrains the statistical power of cross-validation estimates. Class imbalance (62.7% Benign vs. 37.3% Malignant) was not explicitly addressed through oversampling (SMOTE) or cost-sensitive learning, which may have further improved Recall for the Malignant class. Additionally, the study does not evaluate model performance at clinically relevant

operating thresholds (e.g., Recall ≥ 0.99), which is the practically meaningful constraint in cancer screening.

V. Conclusion

This study presented a rigorous comparative evaluation of five machine-learning and deep-learning classifiers for binary breast cancer classification on the Wisconsin Breast Cancer Diagnostic dataset. A standardized preprocessing pipeline — comprising stratified splitting, StandardScaler normalization, and five-fold cross-validation — ensured fair and reproducible model comparison.

Logistic Regression achieved the best overall performance with 98.25% accuracy and an F1-Score of 0.9861, validating its suitability as a high-performance, interpretable baseline for clinical binary classification tasks on small, high-quality tabular datasets. The Deep Neural Network (MLPClassifier, 256-128-64-32 topology) demonstrated the highest ROC-AUC of 0.9960, suggesting superior probability calibration beneficial for adjustable-threshold clinical applications. All five models achieved ROC-AUC values above 0.985, confirming the strong discriminative information embedded in FNA-derived cytological measurements.

Feature importance analysis identified worst concave points, worst radius, and mean concave points as the most diagnostically informative predictors, consistent with established histopathological markers of malignancy. These results support the deployment of Logistic Regression as a computationally efficient, explainable clinical decision-support tool, with the DNN as a complementary probabilistic risk-ranking system.

Future work should investigate: (1) incorporation of class-imbalance correction via SMOTE or class-weighted loss functions to improve Malignant Recall; (2) application of SHAP (SHapley Additive exPlanations) values for model-agnostic interpretability; (3) validation on external datasets from multiple institutions; and (4) integration of imaging features from digital mammography to complement FNA cytology measurements. Federated learning frameworks may enable multi-institutional model training while preserving patient data privacy, a critical consideration for clinical AI adoption.

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